

Exp - 7 Spectral Estimation

7.1 Power Spectral Density Estimation and Spectral Analysis of Noisy Sinusoidal Signals Using Periodogram Method in MATLAB

Program :

```
clc; clear; close all

fs=1000; t=(0:999)/fs;

s=sin(2*pi*100*t)+0.5*sin(2*pi*200*t);

n=0.3*randn(size(t));

x=s+n;

figure, plot(x), grid on

title('Signal with Noise')

figure, periodogram(x,[],[],fs)

title('PSD using Periodogram')

figure

X=abs(fft(x));

f=(0:length(X)-1)*fs/length(X);

plot(f(1:500),X(1:500)), grid on

title('Magnitude Spectrum')

figure

bar([mean(s.^2) var(n)])

set(gca,'XTickLabel',{'Signal','Noise'})

title('Power Comparison')

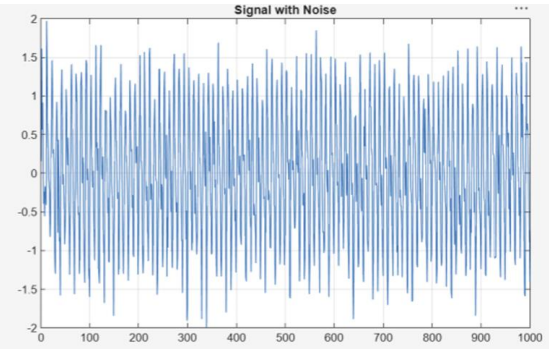
grid on
```

Output :

```

1 clc; clear; close all
2 fs=1000; t=(0:999)/fs;
3 s=sin(2*pi*100*t)+0.5*sin(2*pi*200*t);
4 n=0.3*randn(size(t));
5 x=s+n;
6 figure, plot(x), grid on
7 title('Signal with Noise')
8 figure, periodogram(x,[],1,fs)
9 title('PSD using Periodogram')
10 figure
11 X=abs(fft(x));
12 f=(0:length(X)-1)*fs/length(X);
13 plot(f(1:500),X(1:500)), grid on
14 title('Magnitude Spectrum')
15 figure
16 bar([mean(s.^2) var(n)])
17 set(gca,'XTickLabel',{'Signal','Noise'})
18 title('Power Comparison')
19 grid on
20

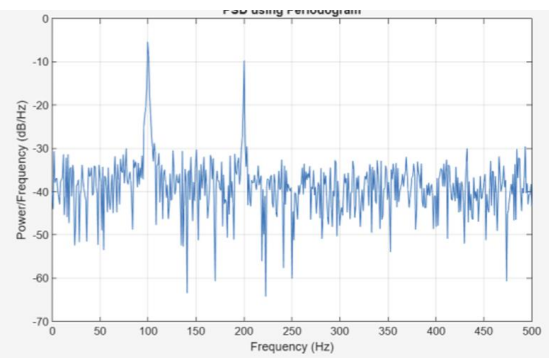
```



```

1 clc; clear; close all
2 fs=1000; t=(0:999)/fs;
3 s=sin(2*pi*100*t)+0.5*sin(2*pi*200*t);
4 n=0.3*randn(size(t));
5 x=s+n;
6 figure, plot(x), grid on
7 title('Signal with Noise')
8 figure, periodogram(x,[],1,fs)
9 title('PSD using Periodogram')
10 figure
11 X=abs(fft(x));
12 f=(0:length(X)-1)*fs/length(X);
13 plot(f(1:500),X(1:500)), grid on
14 title('Magnitude Spectrum')
15 figure
16 bar([mean(s.^2) var(n)])
17 set(gca,'XTickLabel',{'Signal','Noise'})
18 title('Power Comparison')
19 grid on
20

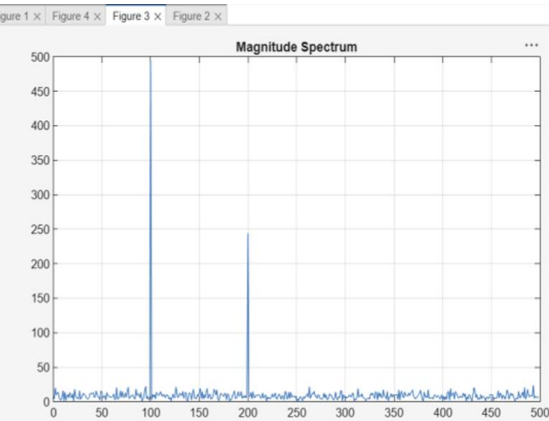
```



```

1 clc; clear; close all
2 fs=1000; t=(0:999)/fs;
3 s=sin(2*pi*100*t)+0.5*sin(2*pi*200*t);
4 n=0.3*randn(size(t));
5 x=s+n;
6 figure, plot(x), grid on
7 title('Signal with Noise')
8 figure, periodogram(x,[],1,fs)
9 title('PSD using Periodogram')
10 figure
11 X=abs(fft(x));
12 f=(0:length(X)-1)*fs/length(X);
13 plot(f(1:500),X(1:500)), grid on
14 title('Magnitude Spectrum')
15 figure
16 bar([mean(s.^2) var(n)])
17 set(gca,'XTickLabel',{'Signal','Noise'})
18 title('Power Comparison')
19 grid on
20

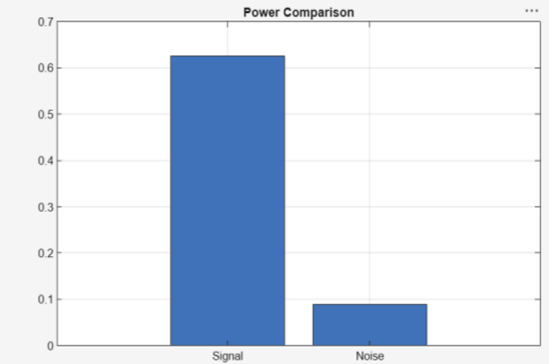
```



```

1 clc; clear; close all
2 fs=1000; t=(0:999)/fs;
3 s=sin(2*pi*100*t)+0.5*sin(2*pi*200*t);
4 n=0.3*randn(size(t));
5 x=s+n;
6 figure, plot(x), grid on
7 title('Signal with Noise')
8 figure, periodogram(x,[],1,fs)
9 title('PSD using Periodogram')
10 figure
11 X=abs(fft(x));
12 f=(0:length(X)-1)*fs/length(X);
13 plot(f(1:500),X(1:500)), grid on
14 title('Magnitude Spectrum')
15 figure
16 bar([mean(s.^2) var(n)])
17 set(gca,'XTickLabel',{'Signal','Noise'})
18 title('Power Comparison')
19 grid on
20

```



7.2 Analysis of Windowing Effects on Spectral Estimation Using Rectangular, Hamming, and Hanning Windows in MATLAB

Program :

```
clc; clear; close all;

% Signal
fs = 1000; N = 256; t = (0:N-1)/fs;
x = sin(2*pi*100*t) + 0.8*sin(2*pi*110*t);
NFFT = 2048; f = (0:NFFT/2-1)*(fs/NFFT);

% Windows
w = {rectwin(N), hamming(N), hann(N)};
names = {'Rectangular','Hamming','Hanning'};

% Processing
for i = 1:3
    xw = x(:).*w{i};
    X = fft(xw,NFFT);
    P{i} = abs(X(1:NFFT/2));
    P{i} = P{i}/max(P{i});
end

%% Individual Spectrums
figure;
for i = 1:3
    subplot(3,1,i);
    plot(f,20*log10(P{i}), 'LineWidth',1.5);
    title(['Spectrum using ' names{i} ' Window']);
    xlim([0 250]); grid on;
end

%% Comparison
figure;
plot(f,20*log10(P{1}), 'k', f,20*log10(P{2}), 'r', f,20*log10(P{3}), 'b', 'LineWidth',1.5);
```

```

legend(names); title('Window Comparison'); xlim([50 170]); grid on;

%%% Window Shapes

figure;

for i = 1:3

subplot(3,1,i); plot(w{i},'LineWidth',1.5);

title([names{i} ' Window']); grid on;

end

%%% Peak Detection

for i = 1:3

[pks{i},loc{i}] = findpeaks(P{i},f,'MinPeakHeight',0.3);

fprintf('\n%s Window Frequencies:\n',names{i});

disp(loc{i});

end

%%% Peak Marking (Hamming)

figure;

plot(f,20*log10(P{2}),'LineWidth',1.5); hold on;

plot(loc{2},20*log10(pks{2}),'ro','MarkerFaceColor','r');

title('Detected Frequencies (Hamming)'); xlim([50 170]); grid on;

%%% Main Peaks

fprintf('\nMain Spectral Peaks:\n');

for i = 1:3

[~,idx] = max(P{i});

fprintf('%s Window Peak = %.2f Hz\n',names{i},f(idx));

end

```

Output:

